

Dataset Report — SoilGrids Soil Features

Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France

SOIL FEATURES

Dataset role: Static parcel-level soil features · **Provider:** ISRIC — SoilGrids v2.0

Access method: WCS raster download (after optimisation from REST API) · **Resolution:** 250 m

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1. Source and provenance

1.1 What is SoilGrids?

SoilGrids v2.0 is a global digital soil-mapping product released by ISRIC — World Soil Information. It uses machine learning (Quantile Random Forest) trained on ~240,000 georeferenced soil profiles worldwide and 400+ environmental covariates (climate, terrain, vegetation, lithology) to predict 13 standard soil properties at six depth intervals on a 250 m global grid.

1.2 Why SoilGrids for this project

Before adopting SoilGrids, three alternative sources were systematically evaluated, with a strong preference for French national programmes producing measured (not modelled) data:

Source	Type	Resolution	Verdict for the project
RMQS (INRAE)	French, real measurements	1 site / 16 km	Too coarse for parcel-level features (~10 sites total in CVL).
BDAT (INRAE)	French, real lab analyses	Aggregated by canton	Better spatial coverage; missing hydraulic properties (FC, WP). Useful as <i>complementary</i> source later.
SoilGrids v2.0	Global, ML-modelled	250 m	Selected. Only source providing parcel-level resolution AND the FC/WP properties needed for water-balance modelling.

Methodological choice: SoilGrids was selected as the parcel-level baseline because no French source provides 250 m resolution combined with hydraulic properties (Field Capacity, Wilting Point). BDAT can later complement SoilGrids by adding canton-level chemistry features (real lab data), in a hybrid approach.

2. Methodological optimisation — REST API vs WCS rasters

SoilGrids exposes two access methods. An initial design used the REST API (one query per parcel), which proved prohibitively slow. A second design used the WCS service to download regional rasters once, then sample values locally. The shift gave a ~91× speed-up **without any loss of data quality**: both methods access the same underlying SoilGrids v2.0 layers.

Method	Queries	Rate limit	Estimated total	Measured total
REST API (per parcel)	1,500	~5/min (polite)	~5 hours	not executed
WCS (per layer)	39	—	~3–5 min	3.3 min

2.1 Why the WCS approach is so much faster — three multipliers

- Far fewer queries (×39 fewer):** the REST API returns one point per call (1,500 calls). The WCS service returns a regional raster per call, so we make one call per layer (39 calls in total).
- No throttling:** the REST API requires polite spacing (~5 req/min). WCS serves pre-rendered raster tiles and has no comparable rate limit.
- Right tool for the use case:** the REST API is built for sparse, point queries. The WCS service is built for bulk regional extraction — exactly our need.

Key observation for the thesis: the optimisation cost nothing in data quality. Both methods access the same SoilGrids v2.0 layers; only the access pattern differs. The measured 91× speed-up illustrates the value of choosing the right tool for the access pattern.

3. Acquisition workflow

Step 1 — Configuration. Loaded the 1,500-parcel sample. Computed the bounding box of Centre-Val de Loire from parcel coordinates, with a small margin ($\sim 0.05^\circ$).

Step 2 — WCS test on a single layer. Downloaded `phh2o_0-5cm` to validate the WCS endpoint and measure the per-layer time (4.0 s for ~ 700 KB). Sanity-checked the mean pH value (6.0) against expected agricultural ranges.

Step 3 — Bulk WCS download. Downloaded the remaining 38 layers (13 properties $\times$ 3 root-zone depths, one layer per combination) — total 43.5 MB, 3.3 minutes, 0 failures.

Step 4 — Per-parcel extraction. Used `rasterio.sample()` to read the raster value under each of the 1,500 parcel centroids — all 39 features for all 1,500 parcels in 3.6 seconds (local I/O).

Step 5 — Quality control. Detected that 2 parcels (0.13%) had value 0 in all 39 features, indicating raster nodata at their exact pixel. These 0 values were replaced with NaN to distinguish them from real data.

Step 6 — Hybrid nearest-neighbour filling. For each of the 2 nodata parcels, the nearest valid pixel was located in each raster (mean distance: 0.22 km = 1 pixel) and its value substituted. Resulting dataset: 100% complete with no NaN.

Step 7 — Export. Saved as `soilgrids_features_cvl_1500.csv` (1,500 $\times$ 45 columns).

4. Key decisions and rationale

Decision: Use WCS rasters instead of the REST API.

Rationale: WCS is designed for regional bulk extraction. With 1,500 parcels concentrated in one region, downloading the raster once is dramatically faster than querying 1,500 points. Same underlying data, $\sim 91\times$ faster.

Decision: Limit depths to the root zone (0–5 cm, 5–15 cm, 15–30 cm).

Rationale: Wheat roots are concentrated in the top 30 cm. Deeper layers (30–60, 60–100, 100–200 cm) are less relevant for harvest-window prediction and would inflate the feature space without gain.

Decision: Retrieve all 13 standard properties (not a sub-set).

Rationale: The download is fast (3 minutes) and complete coverage avoids re-running the workflow if extra features are needed later (e.g. CEC, coarse fragments). Feature selection is deferred to the ML stage where SHAP can identify the truly useful properties.

Decision: Hybrid nearest-neighbour filling rather than dropping the 2 nodata parcels.

Rationale: The substitute pixels are only 1 SoilGrids pixel away (~ 0.22 km), so the imputed values reflect genuinely local soil. Dropping parcels would have been the conservative choice but is unnecessary given the spatial continuity of soil. The approach matches the literature (Selena 2025; McBratney et al., 2003).

5. Deliverable

Item	Value
Output file	<code>soilgrids_features_cvl_1500.csv</code>
Dimensions	1,500 rows $\times$ 45 columns
Identifier columns	6 (<code>parcelle_uid</code> , <code>year</code> , <code>ID_PARCEL</code> , <code>SURF_PARC</code> , <code>lat</code> , <code>lon</code>)
Soil feature columns	39 (13 properties $\times$ 3 depths)

Missing values	0 (after nearest-neighbour filling)
Static raster archive	43.5 MB total, 39 GeoTIFF files in <code>soilgrids_rasters/</code>
Documentation	<code>Data_Dictionary_SoilGrids.pdf</code> (3 pages)

6. Data quality and validation

- **Initial extraction (raw):** 39 values per parcel × 1,500 parcels = 58,500 values. 58,422 valid (99.87%), 78 nodata (2 parcels × 39 layers).
- **After nearest-neighbour filling:** 58,500 valid values (100%).
- **Mean interpolation distance:** 0.22 km — equivalent to 1 SoilGrids pixel.
- **Plausibility checks:** pH mean 6.6 (agricultural range), Field Capacity mean 31%, Wilting Point mean 16% — all physically realistic for the region.
- **Spatial diversity captured:** sand content ranges 5–57%, reflecting the real contrast between Beauce loams and Sologne sandy soils.

7. Relevant observations

- **Static features per parcel:** soil does not change between years, so the same parcel position always yields the same values. The year column tracks the harvest campaign but soil values themselves are time-invariant.
- **Available Water Content (AWC):** can be derived as `wv0033 – wv1500`. This is the agronomically meaningful "plant-available water" feature, very predictive for wheat stress.
- **Optional future enhancement:** BDAT (canton-level real lab measurements) could complement SoilGrids by replacing the modelled chemistry features (pH, N, P, K) with measured values where the parcel falls in a well-sampled canton. Hydraulic features (FC, WP) would still come from SoilGrids.
- **Workflow reusability:** the 39 raster files cover the entire region. Any future re-sampling of parcels (e.g. for other crops) can reuse them without re-downloading.